

A Primer on Optimal Execution

BayQuant

The Trading Dilemma

Liquidating a large position is no free lunch: execution moves prices against the order. Sell it all at once and **market impact** is concentrated into a short, severe shock. Spread it out and impact cost falls, but the remaining shares sit exposed to the market's random swings for longer: **timing risk**. This primer works through the model of Almgren and Chriss [1], who cast optimal liquidation as a mean-variance problem, minimizing a **risk-adjusted cost**: expected cost plus risk aversion times its variance. It derives the optimal schedule and replicates their efficient frontier.

Price Dynamics and Impact

The price follows Arithmetic Brownian Motion (ABM), disturbed by volatility and drift (exogenous) and by the trader's own activity (endogenous). Discretizing the horizon into N steps of length $\tau = T/N$, with grid points $t_k = k\tau$, x_k the signed position still to trade after step k ($x_0 = X$, $x_N = 0$, with $X > 0$ for a sale and $X < 0$ for a purchase; for a liquidation, x_k is simply the shares still held) and $n_k = x_k - x_{k-1}$ the shares traded in step k (positive when buying, negative when selling), the price process is

$$S_k = S_{k-1} + \alpha\tau + \sigma\sqrt{\tau}\xi_k + \tau g\left(\frac{n_k}{\tau}\right), \quad (1)$$

where ξ_k are i.i.d. shocks with zero mean and unit variance, independent of the trading, $g(v)$ is the **permanent impact** (persists in the price process), and the drift α is taken as **zero** throughout since no directional information is assumed.

ABM is an assumption. Its shocks $\sigma\sqrt{\tau}\xi_k$ have a fixed dollar size; under geometric Brownian motion (GBM) they would instead scale with the current price, $\sigma S_{k-1}\sqrt{\tau}\xi_k$, making the risk of each step depend on the path the price has taken. Over trading horizons of days, however, fractional price changes are small, so the two are practically indistinguishable [1]. Moreover, ABM is far more tractable: it keeps the variance of the cost a simple quadratic in the holdings.

Trading also incurs a **temporary impact** $h(v)$: a transient concession on the execution price of a single step,

so the price actually paid (or received, when selling) in step k is

$$\tilde{S}_k = S_{k-1} + h\left(\frac{n_k}{\tau}\right).$$

In both g and h , the argument $v = n_k/\tau$ is the average rate of trading during the interval t_{k-1} to t_k .

Cost of Trading

Following Perold (1988) [2], the **implementation shortfall** $\sum_k n_k (\tilde{S}_k - S_0)$ is the cost of trading relative to the arrival price S_0 : what the trades actually paid, net of what they received, minus what they would have at S_0 . Expanding it gives the shortfall's expectation and variance

$$\begin{aligned} E(x) &= -\sum_k \tau x_k g\left(\frac{n_k}{\tau}\right) + \sum_k n_k h\left(\frac{n_k}{\tau}\right), \\ V(x) &= \sigma^2 \sum_k \tau x_k^2. \end{aligned} \quad (2)$$

Specializing to linear impact, $g(v) = \gamma v$ and $h(v) = \varepsilon \operatorname{sgn}(v) + \eta v$ (with ε a fixed cost per trade, e.g. half the bid-ask spread), summation by parts gives the permanent-impact term

$$-\sum_k \tau x_k g\left(\frac{n_k}{\tau}\right) = \frac{1}{2}\gamma X^2 - \frac{1}{2}\gamma \sum_k n_k^2, \quad (3)$$

so its contribution to $E(x)$ is **not** schedule-independent at finite τ . Temporary impact contributes

$$\sum_k n_k h\left(\frac{n_k}{\tau}\right) = \varepsilon \sum_k |n_k| + (\eta/\tau) \sum_k n_k^2.$$

Combining, and writing $\bar{\eta} = \eta - \frac{1}{2}\gamma\tau$ to absorb the schedule-dependent permanent-impact term,

$$E(x) = \frac{1}{2}\gamma X^2 + \varepsilon \sum_k |n_k| + \frac{\bar{\eta}}{\tau} \sum_k n_k^2.$$

For a monotone schedule (n_k all one sign), $\sum_k |n_k| = |X|$ (a telescoping sum), so only the last term shapes the schedule. Almgren and Chriss minimize the utility function

$$U(x) = E(x) + \lambda V(x)$$

over $x_1, \dots, x_{N-1}$, where $\lambda \geq 0$ is risk aversion.

The Optimal Schedule

The terms $\frac{1}{2}\gamma X^2 + \varepsilon|X|$ are the same for every schedule, so they do not change the optimal schedule and can be dropped. What remains is a quadratic in the position still to trade,

$$U(x) = \frac{\bar{\eta}}{\tau} \sum_k (x_k - x_{k-1})^2 + \lambda \sigma^2 \tau \sum_k x_k^2,$$

with the endpoints $x_0 = X$ and $x_N = 0$ held fixed and only the interior positions $x_1, \dots, x_{N-1}$ free. The first sum penalizes trading fast (impact cost), the second penalizes holding inventory (timing risk). $U(x)$ is convex, so its unique minimum is where every partial derivative vanishes, which gives the second-order linear recurrence

$$\frac{x_{j+1} - 2x_j + x_{j-1}}{\tau^2} = \tilde{\kappa}^2 x_j, \quad \tilde{\kappa}^2 = \frac{\lambda \sigma^2}{\bar{\eta}}, \quad (4)$$

a discrete analogue of $\ddot{x}(t) = \tilde{\kappa}^2 x(t)$, where:

- $x(t)$ is the position still to trade at time t ;
- $\ddot{x}(t)$ measures how the pace of trading $|\dot{x}(t)|$ changes: zero at a constant pace, positive when a seller slows down;
- $\tilde{\kappa}^2$ is the price of risk relative to the price of impact.

Multiplying by $\bar{\eta}$ gives $\bar{\eta}\ddot{x}(t) = \lambda\sigma^2 x(t)$: at every moment, the change in the marginal impact cost of trading balances the marginal risk cost of the $x(t)$ shares still to trade. The trader therefore trades fastest while that position is large and slows down as it shrinks, so the path bows toward zero from the straight line of constant-rate trading: convex for a sale.

Solving it gives a decay rate θ , the solution of

$$\frac{2}{\tau^2}(\cosh(\theta\tau) - 1) = \tilde{\kappa}^2, \quad (5)$$

and the optimal liquidation trajectory

$$x_j = X \frac{\sinh(\theta(T - t_j))}{\sinh(\theta T)}, \quad (6)$$

for $j = 0, \dots, N$, with the associated trade list

$$n_j = -\frac{2 \sinh(\frac{1}{2}\theta\tau)}{\sinh(\theta T)} \cosh(\theta(T - t_{j-\frac{1}{2}}))X, \quad (7)$$

for $j = 1, \dots, N$, where $t_{j-\frac{1}{2}} = (j - \frac{1}{2})\tau$ is the midpoint of step j .

Continuous-time limit. As $\tau \rightarrow 0$, $\bar{\eta} \rightarrow \eta$ and both θ and $\tilde{\kappa}$ tend to $\kappa = \sqrt{\lambda\sigma^2/\eta}$. The trajectory (6) becomes

$$x(t) = X \frac{\sinh(\kappa(T - t))}{\sinh(\kappa T)},$$

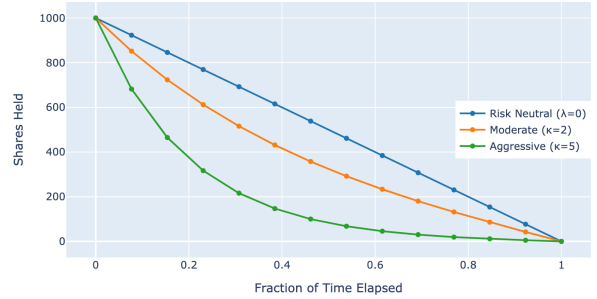
the solution of $\ddot{x}(t) = \kappa^2 x(t)$, and the trading rate n_j/τ from (7) becomes

$$\dot{x}(t) = -X\kappa \frac{\cosh(\kappa(T - t))}{\sinh(\kappa T)}.$$

The Shape of the Schedule

θ alone governs the curve's shape. As $\lambda \rightarrow 0$, $\theta \rightarrow 0$ and $\sinh(z) \approx z$ makes the schedule linear: constant-rate, TWAP-like execution. As λ grows, θ grows and the curve front-loads, accepting more impact cost in return for less timing risk.

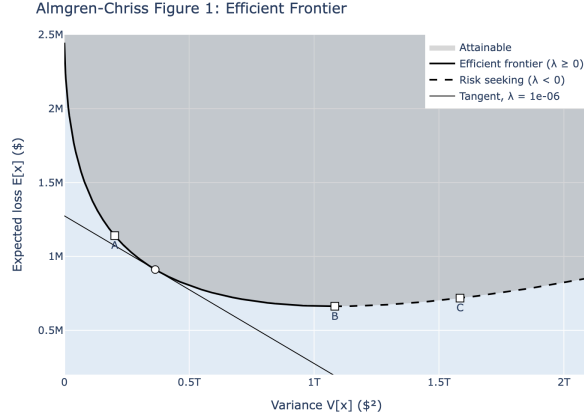
Optimal Holdings Trajectory x_j



The Efficient Frontier

Sweeping $\lambda \geq 0$ traces out a curve of $(V(x), E(x))$ pairs, one per optimal trajectory: the **efficient frontier** of optimal execution. Each point is the lowest expected cost achievable at that variance; no monotone schedule beats the frontier on both at once. At $\lambda = 0$ the frontier sits at its minimum-cost, maximum-variance end (the TWAP schedule, which minimizes $\sum_k n_k^2$ for fixed X); increasing λ moves along the frontier toward lower variance at higher cost as the schedule front-loads.

The figure replicates Figure 1 of [1] with their test-case parameters ($X = 10^6$ shares, $T = 5$ days, $N = 5$). Past the risk-neutral point B, $\lambda < 0$ (dashed) describes a risk-seeking trader who delays selling and pays more in both cost and variance, so that branch is not efficient. At each point, the slope of the frontier is $-\lambda$; the straight line shows this tangent at $\lambda = 10^{-6}$.



Half-Life of a Trade

For $\theta T \gtrsim 1$, $\sinh(\theta(T-t))/\sinh(\theta T) \approx e^{-\theta t}$, so holdings decay approximately exponentially, $x(t) \approx X e^{-\theta t}$, and $1/\theta$ is the timescale of the unwind: after a time $1/\theta$, about $e^{-1} \approx 37\%$ of the position remains. The time for the position to fall to half its initial size is the **half-life**

$$t_{1/2} = \frac{\ln(2)}{\theta} \approx \frac{\ln(2)}{\kappa} = \ln(2) \sqrt{\frac{\eta}{\lambda \sigma^2}}.$$

Half-life shrinks with risk aversion λ and volatility σ (faster unwinds when risk is costlier) and grows with the temporary-impact coefficient η (slower unwinds when trading is more expensive).

Buying as the Mirror Image

A purchase is the same problem with $X < 0$: the position still to trade starts at $x_0 = X$ and rises to zero, so (6) and (7) apply unchanged and every trade n_j is positive. The risk sits on the shares not yet bought, so a risk-averse buyer, like a seller, trades fastest at the start. The shares held, $|X| + x_j$, follow

$$|X| \left(1 - \frac{\sinh(\theta(T-t_j))}{\sinh(\theta T)} \right),$$

a concave path that rises fastest early on.

Implementation: Numerical Stability

Equations (5) to (7) are exact but fail in floating point at both ends of the risk-aversion range. Three rewrites keep them accurate.

Decay rate. Inverting (5) directly gives $\theta = \tau^{-1} \operatorname{arccosh}(1 + \frac{1}{2} \tilde{\kappa}^2 \tau^2)$, but for small $\tilde{\kappa} \tau$ the argument rounds to 1 and θ collapses to 0 (in double precision already at $\tilde{\kappa} \tau \approx 10^{-8}$). Since $\cosh z - 1 =$

$2 \sinh^2(z/2)$, (5) is equivalent to $2\tau^{-1} \sinh(\frac{1}{2}\theta\tau) = \tilde{\kappa}$, which inverts without cancellation:

$$\theta = \frac{2}{\tau} \operatorname{arcsinh}\left(\frac{\tilde{\kappa}\tau}{2}\right).$$

Overflow. $\sinh(\theta T)$ overflows in double precision once $\theta T \gtrsim 710$, and (6) and (7) then evaluate to ∞/∞ . Factoring out the dominant exponential leaves only non-positive exponents,

$$x_j = X e^{-\theta t_j} \frac{1 - e^{-2\theta(T-t_j)}}{1 - e^{-2\theta T}},$$

$$n_j = -X \frac{2 \sinh(\frac{1}{2}\theta\tau) e^{-\theta t_{j-\frac{1}{2}}} \left(1 + e^{-2\theta(T-t_{j-\frac{1}{2}})} \right)}{1 - e^{-2\theta T}}.$$

Computing the trades from (7) rather than as $x_j - x_{j-1}$ also avoids cancellation late in an aggressive schedule, when consecutive holdings are nearly equal.

Risk-neutral limit. As $\theta \rightarrow 0$, each $1 - e^{-u}$ factor above loses its leading digits to cancellation; evaluating it as $-\operatorname{expm1}(-u)$ keeps full precision for small u . At $\theta = 0$ exactly the expressions are 0/0, so that case returns the linear schedule $x_j = X(T-t_j)/T$, $n_j = -X/N$ directly.

Appendix

Price Process from Continuous Time

Equation (1)

$$dS_t = (\alpha + g(v_t)) dt + \sigma dW_t,$$

where v_t is the trader's continuous trading rate (positive when buying). Discretizing via Euler-Maruyama, with n_k/τ standing in for v_t and $W_{t_k} - W_{t_{k-1}} \approx \sqrt{\tau}\xi_k$ for the Brownian increment over step k , gives

$$S_k = S_{k-1} + \alpha\tau + \sigma\sqrt{\tau}\xi_k + \tau g\left(\frac{n_k}{\tau}\right), \quad k = 1, \dots, N.$$

Shortfall Identity

Equation (2)

Temporary impact. Substituting the execution price $\tilde{S}_k = S_{k-1} + h\left(\frac{n_k}{\tau}\right)$ splits the shortfall into trading at the prevailing price plus the temporary-impact concession,

$$\sum_{k=1}^N n_k (\tilde{S}_k - S_0) = \sum_{k=1}^N n_k (S_{k-1} - S_0) + \sum_{k=1}^N n_k h\left(\frac{n_k}{\tau}\right).$$

Summation by parts. Writing $S_{k-1} - S_0$ as the sum of the price changes $S_i - S_{i-1}$ over $i < k$ and swapping the order of summation,

$$\begin{aligned} \sum_{k=1}^N n_k (S_{k-1} - S_0) &= \sum_{i=1}^{N-1} (S_i - S_{i-1}) \sum_{k=i+1}^N n_k \\ &= - \sum_{i=1}^N x_i (S_i - S_{i-1}), \end{aligned}$$

since the trades after step i add up to $x_N - x_i = -x_i$ (the $i = N$ term is zero because $x_N = 0$). Each price change counts against the position still to trade: a rise lowers a seller's cost ($x_i > 0$) and raises a buyer's ($x_i < 0$).

Price dynamics. By equation (1) with zero drift, each price change is $S_k - S_{k-1} = \sigma\sqrt{\tau}\xi_k + \tau g\left(\frac{n_k}{\tau}\right)$. Substituting it gives the shortfall identity

$$\sum_{k=1}^N n_k (\tilde{S}_k - S_0) = - \sum_{k=1}^N \sigma\sqrt{\tau}\xi_k x_k - \sum_{k=1}^N \tau x_k g\left(\frac{n_k}{\tau}\right) + \sum_{k=1}^N n_k h\left(\frac{n_k}{\tau}\right).$$

Expectation and variance. Only the first sum is random. The ξ_k have zero mean, so it drops out of the expectation, giving $E(x)$; the last two sums are fixed by the schedule, and the ξ_k are independent with unit variance, so the variance is $V(x) = \sigma^2 \sum_k \tau x_k^2$.

Permanent-Impact Summation by Parts

Equation (3)

Since $n_k = x_k - x_{k-1}$,

$$x_k^2 - x_{k-1}^2 = (x_k - x_{k-1})(x_k + x_{k-1}) = n_k x_k + n_k x_{k-1}.$$

Summing over $k = 1, \dots, N$ telescopes the left side to $x_N^2 - x_0^2 = -X^2$, giving

$$\sum_k n_k x_k + \sum_k n_k x_{k-1} = -X^2.$$

Separately, $n_{k(x_k - x_{k-1})} = n_k^2$, so

$$\sum_k n_k x_k - \sum_k n_k x_{k-1} = \sum_k n_k^2.$$

Adding the two identities and dividing by 2,

$$\sum_k n_k x_k = -\frac{1}{2}X^2 + \frac{1}{2}\sum_k n_k^2.$$

With $\tau x_k g(\frac{n_k}{\tau}) = \gamma n_k x_k$, multiplying by $-\gamma$ gives equation (3).

Optimal-Schedule Recurrence

Equation (4)

Take general endpoints $x_0 = X_0$ and $x_N = X_T$. $U(x)$ is a sum of squares, a convex bowl with a single minimum where every partial derivative vanishes. For an interior x_j , the only terms containing it are $(x_j - x_{j-1})^2$, $(x_{j+1} - x_j)^2$, and $\lambda \sigma^2 \tau x_j^2$, so

$$\frac{\partial U}{\partial x_j} = \frac{\bar{\eta}}{\tau} [-2(x_{j-1} - x_j) + 2(x_j - x_{j+1})] + 2\lambda \sigma^2 \tau x_j = 0.$$

Dividing by $2\bar{\eta}/\tau$ and rearranging gives

$$x_{j+1} - 2x_j + x_{j-1} = \frac{\lambda \sigma^2}{\bar{\eta}} \tau^2 x_j = \tilde{\kappa}^2 \tau^2 x_j,$$

and dividing by τ^2 gives equation (4).

Decay Rate

Equation (5)

Equation (4) is a second-order linear homogeneous difference equation with constant coefficients and values fixed at both ends. Its solution space is two-dimensional, so the task is to find two independent solutions and fit two constants.

Substituting the trial solution $x_j = e^{\theta t_j}$, with $t_{j\pm 1} = t_j \pm \tau$,

$$\frac{e^{\theta\tau} - 2 + e^{-\theta\tau}}{\tau^2} e^{\theta t_j} = \frac{2}{\tau^2} (\cosh(\theta\tau) - 1) e^{\theta t_j} = \tilde{\kappa}^2 e^{\theta t_j}.$$

The factor $e^{\theta t_j}$ never vanishes, so $e^{\theta t_j}$ is an exact solution whenever θ satisfies the cosh relation, equation (5). θ differs from $\tilde{\kappa}$ because a second difference is not a second derivative: $2(\cosh z - 1) = z^2 + z^4/12 + \dots$, so $\theta = \tilde{\kappa} + O(\tau^2)$. Because cosh is even, $e^{-\theta t_j}$ is also a solution; the ratio $e^{2\theta t_j}$ of the two is non-constant for $\theta > 0$, so they are independent and span every solution.

Optimal Liquidation Trajectory

Equation (6)

Boundary conditions. The general solution of equation (4) is

$$x_j = A e^{\theta t_j} + B e^{-\theta t_j},$$

and the two endpoints give a linear system in A and B ,

$$A + B = X_0, \quad A e^{\theta T} + B e^{-\theta T} = X_T.$$

From the first equation $B = X_0 - A$; substituting into the second and using $e^{\theta T} - e^{-\theta T} = 2 \sinh(\theta T)$,

$$A = \frac{X_T - X_0 e^{-\theta T}}{2 \sinh(\theta T)}, \quad B = \frac{X_0 e^{\theta T} - X_T}{2 \sinh(\theta T)}.$$

Reassembling $x_j = A e^{\theta t_j} + B e^{-\theta t_j}$ and grouping the two coefficients,

$$\begin{aligned} 2 \sinh(\theta T) x_j &= X_T (e^{\theta t_j} - e^{-\theta t_j}) + X_0 (e^{\theta T} e^{-\theta t_j} - e^{-\theta T} e^{\theta t_j}) \\ &= X_T (e^{\theta t_j} - e^{-\theta t_j}) + X_0 (e^{\theta(T-t_j)} - e^{-\theta(T-t_j)}) \\ &= 2X_T \sinh(\theta t_j) + 2X_0 \sinh(\theta(T-t_j)), \end{aligned}$$

where the second line used $e^{\theta T} e^{-\theta t_j} = e^{\theta(T-t_j)}$ and $e^{-\theta T} e^{\theta t_j} = e^{-\theta(T-t_j)}$. Each $\sinh$ emerges on its own from a pair of exponentials of opposite sign, so no basis has to be guessed in advance. Dividing by $2 \sinh(\theta T)$ gives

$$x_j = \frac{X_0 \sinh(\theta(T-t_j)) + X_T \sinh(\theta t_j)}{\sinh(\theta T)},$$

a hyperbolic interpolation between the endpoints: the first term carries X_0 and vanishes at $t = T$, the second carries X_T and vanishes at $t = 0$. As $\theta \rightarrow 0$ it reduces to the straight line $x_j = (X_0(T-t_j) + X_T t_j)/T$, and larger θ bows the path further from it. Setting $X_0 = X$, $X_T = 0$ gives equation (6).

Time consistency. Re-planning at step m from the current position x_m , with $X_T = 0$ over the remaining horizon $T - t_m$, gives

$$x_j = x_m \frac{\sinh(\theta(T-t_j))}{\sinh(\theta(T-t_m))}, \quad j = m, \dots, N.$$

If x_m is on schedule, $x_m = X \sinh(\theta(T-t_m))/\sinh(\theta T)$ by equation (6), the factor $\sinh(\theta(T-t_m))$ cancels and the re-planned path is exactly the remainder of equation (6).

Trade List

Equation (7)

Since $n_j = x_j - x_{j-1}$, equation (6) gives

$$n_j = -\frac{X}{\sinh(\theta T)} [\sinh(\theta(T-t_{j-1})) - \sinh(\theta(T-t_j))].$$

The identity $\sinh u - \sinh v = 2 \cosh((u+v)/2) \sinh((u-v)/2)$, with $u-v = \theta\tau$ and $(u+v)/2 = \theta(T-t_{j-\frac{1}{2}})$, gives equation (7).

Notation

symbol	meaning	units
X	signed size of the program: shares to liquidate ($X > 0$), or minus the shares to accumulate ($X < 0$)	shares
X_0, X_T	position still to trade at the start and end of a schedule; $X_0 = X$ and $X_T = 0$ for a full program	shares
T	trading horizon	day
$\tau = T/N$	length of one discretization step, with N the number of steps	day
$x_k, x(t)$	signed position still to trade after step k , or at time t ; the shares still held, for a liquidation	shares
$S_k, S(t)$	security price after step k , or at time t , with S_0 the initial price	\$/share

symbol	meaning	units
α	drift rate of the price process; assumed zero throughout	\$/share/day
ξ_k	i.i.d. shock in step k , zero mean and unit variance	—
$n_k = x_k - x_{k-1}$	shares traded in step k : positive when buying, negative when selling	shares
n_k/τ	trading rate during step k : shares traded per unit time, not to be confused with n_k itself	shares/day
$g(v)$	function of trading rate v entering the price process; its effect carries forward into S_k	\$/share
$\tilde{S}_k$	actual price per share paid (or received, when selling) in step k	\$/share
$h(v)$	temporary-impact function: the concession on the average price per share from trading at rate v during one interval, paid above the price when buying and received below it when selling; does not carry forward into S_k	\$/share
$E(x)$	expected cost of trading schedule x	\$
$V(x)$	variance of the cost of trading schedule x	$\2
$U(x) = E(x) + \lambda V(x)$	utility function minimized over the schedule; convex, so its unique minimum is found by setting $\partial U/\partial x_j = 0$	\$
γ	permanent-impact coefficient: the persistent price shift per share traded	\$/share ²
η	temporary-impact coefficient: the price concession per unit trading rate	\$/share ²
ε	fixed cost per trade in $h(v)$, e.g. half the bid-ask spread plus fees	\$/share
$\bar{\eta} = \eta - \frac{1}{2}\gamma\tau$	temporary-impact coefficient net of the permanent-impact schedule-dependent term; $\bar{\eta} \rightarrow \eta$ as $\tau \rightarrow 0$	\$/share ²
σ	volatility of the traded asset	\$/share/ $\sqrt{\text{day}}$
λ	risk aversion: the trader's chosen weight on cost variance	1/\$
$\tilde{\kappa} = \sqrt{\lambda\sigma^2/\bar{\eta}}$	coefficient of the discrete optimality recurrence	1/day
θ	decay rate of the optimal schedule, solving $2\tau^{-2}(\cosh(\theta\tau) - 1) = \tilde{\kappa}^2$; $\theta \rightarrow \kappa$ as $\tau \rightarrow 0$	1/day
$\kappa = \sqrt{\lambda\sigma^2/\eta}$	continuous-time limit of both θ and $\tilde{\kappa}$	1/day

γ and η each carry an extra factor of 1/ share beyond what “a price shift per share traded” suggests, because **price** is itself quoted in \$/share: a shift in that per-share price, per share traded, is (\$/share)/share = \$/share².

References

- [1] R. Almgren and N. Chriss, “Optimal execution of portfolio transactions,” *Journal of Risk*, vol. 3, no. 2, pp. 5–39, 2000.
- [2] A. F. Perold, “The implementation shortfall: Paper versus reality,” *Journal of Portfolio Management*, vol. 14, no. 3, pp. 4–9, 1988.